THE MAJOR ISSUES IN THE "AGREED" PRINCIPLES ON REMOTE SENSING

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Foreword

In the United Nations Committee on Peaceful Uses of Outer Space (COPUOS), Canada noted, regarding remote sensing, that there is no automaticity in the application of international law to this new activity.[1] Professor Bin Cheng in a brilliant article denies this statement but adds: "What is true is that there is not a specific international legal regime governing remote sensing". This remark was made in March 1978[2] and still stands; almost no progress has been made since the 1978 version of the "Draft Principles on Remote Sensing of Earth from Space." The Working Group has agreed only to remove brackets or shift the sequence of the article. Some guidelines, however, have acquired the consensus of the delegates to the Legal Subcommittee of COPUOS. With time and the effective operation of remote sensing a general acceptance may be reached. Meanwhile, the two major issues are still discussed: the permissibility of sensing other States' territory and the dissemination of the resultant data and information.

I. Introduction

With the energy crisis, man realized the limited nature of global natural resources and the necessity of preserving the environment of our planet. It has not been surprising that those space technologies employed in the late 1950's and 1960's for purposes of military reconnaissance and meteorology have been extended to the study of the natural resources of the earth and its environment. This technology is termed remote sensing, a "system of methods for identifying the nature and for determining the condition of objects on the earth's surface and of phenomena on, below or above it, by means of observations from airborne or spaceborne platforms."[3] This paper will focus on the legal issues presented by remote sensing activities which, while conducted in space, are earth-oriented.

The sensing techniques employed on satellites take many forms. Besides sophisticated camera systems, infrared, microwave and ultraviolet sensors, as well as radar and laser instruments, are in common use.[4] Sensing can be either passive or active.

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[1]U.N. COPUOS Doc. A/AC.105/C.1/WG.4 L.11, para. 25 (1974).

[2]Cheng, "Legal Implications of Remote Sensing," Proceedings of an International Conference on Earth Observation from Space and Management of Planetary Resources, held at Toulouse, March 6-11, 1978 (ESA SP-134).

[3]U.N. Doc. A/AC. 105/98 (1972).

[4]Christol, "Remote Sensing and International Law", V Annals of Air and Space Law, 375 at 379 (1980).

Passive sensing involves the reading of the natural radiation emitted by an object. Active sensing, on the other hand, is accomplished by first transmitting electromagnetic radiation down to the object and then reading the reflected energy. Since each object emits differing amounts of electromagnetic radiation, it is possible to obtain its distinctive "spectral signature". This energy distribution differentiates the object from others and can indicate density, surface regularity, moisture content and various other physical and chemical properties.[5] Comparisons of the data collected in space are made with measurements taken of the object at the surface of the earth for correlation purposes. Over the years, sensing equipment has been refined to the point that objects on the ground can be identified with precision down to less than 10 meters.[6]

Remote sensing serves many functional purposes. Information can be gathered relating to the environment, agriculture and forestry, geography, geology and mineral resources, oceanography and marine resources, hydrology and water resources, and atmosphere-meteorology, including disaster assessment.[7] The environmental, economic, social, cultural and scientific benefits to be gained are obvious. As paradoxical as it may sound, through military surveillance satellites, remote sensing also serves to promote international peace and stability by providing a means of weapon verification.[8]

Any successful remote sensing system must contain three elements: (1) one or more satellites; (2) a ground data handling center; and (3) an adequate infrastructure, including a good interpretation and analysis capability and an effective structure for the making and carrying out of policy decisions.[9] The American LANDSAT Program is a prime example of both the versatility of sensing techniques and the organizational framework necessary for national and international benefits. LANDSATS 1 and 2, launched in 1972 and 1975 respectively, have already returned to earth after providing information on types and conditions of vegetation, surface, soil and rock conditions, water quality, and other information beneficial to man's needs.[10] LANDSAT 3, launched in 1978, is still functional and a fourth satellite is to be put into operation in 1982. LANDSAT data has been distributed to over 100 countries and there are 8 ground stations in 7 countries outside the United States. Bilateral negotiations are continuing for ground stations to be built in at least 6 other nations, including China. In Europe, ESA (European Space Agency) is planning its own sensing satellite and France alone is developing its Spot satellite for the same purpose.

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[5]Doyle, "Remote Sensing by Satellite: Technical and Operational Implications for International Cooperation" in N. M. Matte and H. DeSaussure (eds.), *Legal Implications of Remote Sensing From Outer Space* 5 (1976).

[6]Christol, *supra* note 4, at 380.

[7]U.N. Doc. A/AC.105/118, at 32-34 (1973).

[8]Christol, *supra* note 4, at 379.

[9]Henein, "Notes on the 'Real World' Framework for Space Law as Applied to Remote Sensing" in N. M. Matte and H. DeSaussure (eds.), *supra* note 5, at 141-142.

[10]Doyle, *supra* note 5, at 6-7.

Remote sensing has not developed without legal consideration. The United Nations Committee on the Peaceful Uses of Outer Space (COPUOS) has been the forum of most of these efforts.[11] Substantive discussions took place in Vienna in 1968 at the first U.N. Conference on the Peaceful Uses of Outer Space. The Scientific and Technical Subcommittee called attention to the use of remote sensing techniques as a means suited to the planning of global resources in 1969. In 1971, an interdisciplinary "Working Group on Remote Sensing of the Earth by Satellites" came into being as an organ within the Scientific and Technical Subcommittee and in 1974, the subject of the legal implications of remote sensing from space was placed on the agenda of the Legal Subcommittee. By this time, 7 proposals of draft principles were in existence: Argentina (1970);[12] France (1969);[13] the USSR (1973);[14] Brazil (1974);[15] France/U.S.S.R. (1974);[16] Argentina/Brazil (1974);[17] and USA (1975).[18] Working with these drafts, as well as input from other members of COPUOS, the Working Group of the Legal Subcommittee was able to identify 5 common principles in 1975.[19] In 1979 the number of principles stood at 17,[20] many, however, without consensus and subject to alternate formulations.

II. The Rights and Obligations of Sensing and Sensed States

A. Introduction

In 1976 the Scientific and Technical Subcommittee of COPUOS identified the structural aspects of remote sensing to include:

(1) Date acquisition (satellites and command stations);
(2) Data reception (antennas and receivers);
(3) Data pre-processing (formatting and recording);
(4) Data storage and dissemination (archiving and reproduction);

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[11]*See* Vlasic, "The Evolution of the International Code of Conduct to Govern Remote Sensing by Satellite Progress Report," 3 Ann. Air & Space L. 561-562 (1978).

[12]U.N. Doc. A/AC.105/C.2/L.73 (1970).

[13]U.N. Doc. A/AC.105/C.2/L.69 (1969).

[14]U.N. Doc. A/AC.105/C.2/L.88 (1973).

[15]U.N. Doc. A/AC.105/122 (1974).

[16]U.N. Doc. A/AC.105/C.2, L.99 (1974).

[17]U.N. Doc. A/AC.105/C.1, 1047 (1974).

[18]U.N. Doc. A/AC.105/C.2, L.103 (1975).

[19]U.N. Doc. A/AC.105/147, Annex 3 at 1-2 (1975).

[20]U.N. Doc. A/AC.105/240, Annex 1 at 7 (1979) (hereinafter referred to as "1979 Principles"), reproduced in 8 J. Space L. 147-149 (1980).

(5) Data analysis (interpretation or user processing);
(6) Information utilization (practical applications by users).[21]

The first four elements concern what is referred to today as "primary data" and which is defined as

> [T]hose primary data which are acquired by satellite-borne sensors and transmitted from a satellite either by telemetry in the form of electromagnetic signals or physically in any form such as photographic film or magnetic tape, as well as preprocessed products derived from those data which may be used for later analysis.[22]

The final two elements concern "analyzed information" which is the "end product resulting from the analytical process performed on the primary data combined with data and/or knowledge obtained from sources other than satellite-borne remote sensors."[23]

The 1979 Principles also define "remote sensing of the earth" as meaning "remote sensing of the earth and its environment."[24] While all three of the above definitions are subject to further debate, they serve as a starting point for this discussion. The major issues are: the right to sense, *ie,* the right of a State to acquire primary data of another State and the right of the sensing State to disseminate both the primary data and the analyzed information concerning the resources of the sensed State.

B. The Right to Sense

The right to sense has been the most controversial issue throughout the discussions in COPUOS. By 1979 a number of positions as to the legality and conditions under which remote sensing might take place had been articulated. These positions ranged from the uninhibited right to sense the resources of another State to one of a prohibition without the prior consent of the sensed State. The fundamental principle behind all this was that of sovereignty, with all of its political, economic and legal trappings. The sovereignty of the sensing State to participate freely in outer space activities confronted the sovereignty of the sensed State over its natural resources and, as advocated by some States, the information pertaining thereto. Between these two extremes were those States which accepted the legality and necessity of remote sensing in a general way and sought only prior notification or consultation regarding the sensing of their territory and those States which were more concerned with the dissemination of the primary data and analyzed information to third parties.[25]

Those States which would require prior consent as a pre-condition for the sensing of their territory based their arguments on territorial sovereignty, sovereignty over natural resources and wealth, and the security and economic considerations therein involved.

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[21]U.N. Doc. A/AC.105/170 at 8 (1976).

[22]1979 Principles, *supra* n. 20, Principle I (b).

[23]1979 Principles, *supra* n. 20, Principle I (c).

[24]1979 Principles, *supra* n. 20, Principle I (a).

[25]Christol, *supra* note 4, at 392-397; *see also* Cheng, *supra* note 2, at para. 2.1.

This position was put forth in the joint Argentina/Brazil Draft Treaty of 1974.[26] Article V of that Draft established the strict rule that States shall refrain from undertaking activities of remote sensing of natural resources belonging to another State without the consent of that State. Article VI of the same Draft provided that States will take all measures authorized by international law to protect their territory from remote sensing activities for which they had denied their consent. It is unclear whether this provision would extend to the destruction of the offending satellite by the sensed State.

Not unexpectedly, the United States has been the main proponent of the position advocating complete freedom to sense. It should be stated at the outset that this is the preferable view not only legally but also economically, socially and technologically. The United States has always maintained that the free collection of data is in accordance with international law.[27] Although earth-oriented, remote sensing is a space activity and, therefore, subject to the 1967 Principles Treaty and its fundamental principle that outer space is free for the exploration and use by all States. In the 1973 United Nations Secretariat Background Paper entitled "Legal Implications of Remote Sensing of the Earth by Satellites", it was stated: (1) there does not appear to be any principle or rule of international law that makes it unlawful for a country to freely observe everything and anything in another country so long as it carries out its observations from beyond the limits of national sovereignty; (2) the only restrictions are those contained in the obligation to act in accordance with international law and to respect the corresponding interest of other States, as well as the duty to inform the United Nations Secretary General and the public, to the greatest extent feasible and practicable, of the nature, conduct, locations and results of national space activities.[28] One can also point to customary international law which has evolved over the last decade from the established practice of States in conducting peaceful space activities.[29]

The United States' position was reiterated in the Presidential Directive of June 20, 1978, which provided that: (1) the United States rejects any claims to the sovereignty over outer space or over celestial bodies, or any portion thereof, and rejects any limitations on the fundamental right to acquire data from space; and (2) the United States will develop and operate on a global basis active and passive remote sensing activities in support of national objectives.

The right or freedom to sense is perhaps best justified on other than legal grounds. From the technical aspect, satellites are not able, despite their scientific sophistication to detect invisible political boundaries.[30] Such determinations would be necessary if the right to sense was premised on the prior consent of the many States scanned by the satellite. It is also clear that the problems to which remote sensing are addressed are of

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[26]*See* n. 17 *supra*.

[27]*See* Christol, *supra* note 4, at 395.

[28]U.N. Doc. A/AC.105/118 (1973).

[29]Christol, *supra* note 4, at 393.

[30]Leigh, "United States Policy of Collecting and Disseminating Remote Sensing Data," in N. M. Matte and H. DeSaussure (eds.), *supra* note 5, at 149.

global, or at least regional, proportion.[31] An open system of sensing and distribution of information would contribute to the preservation of the environment and effective management and control by States of their natural resources. The United States has long advocated international cooperation in these endeavors. These principles will be further developed when the right to disseminate primary data and analyzed information is discussed.

The Legal Subcommittee of COPUOS has been trying to formulate a legal framework for remote sensing activities since 1975. Many of the 1979 Principles, of which are seventeen, simply restate those principles found in such documents as the Principles Treaty, the Registration Convention, the Charter of the United Nations, the UN Declaration on Principles of International Law concerning Friendly Relations as well as general principles of international law.[32] The rights and duties of States are often articulated in the broadest of terms or are ambiguous. It should also be noted that those principles which are regarded as in derogation of the sovereignty of either the sensing or the sensed State are without consensus. All of the Principles, however, are based on the assumption that States have a right to engage in remote sensing of the earth and its environment.[33] Those principles which seek to impose specific conditions on the collection of data and dissemination of data and information do provide a useful starting point from which to establish a legal regime for remote sensing activities.

Areas of general agreement in the 1979 Principles include: the distinction between primary data and analyzed information;[34] the fact that sensing is to be carried out for the benefit and in the interest of all countries (but with special consideration for the needs of the developing countries);[35] and the understanding that sensing is to be subject to international law, the U.N. Charter and the 1967 Principles Treaty.[36] Frequent references are made to international cooperation.[37]

With respect to the right to sense, it is provided that a State which intends to conduct remote sensing of the earth from outer space shall give "advance notification" to the States whose territory will be sensed.[38] The Secretary General of the United Nations is also to receive notice of the sensing activities and publish such information received.[39] This is a much more liberal regime than that of "prior consent". In fact, it cannot be regarded as a serious abridgment of the freedom to conduct remote sensing of

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[31] *Id.*

[32] Vlasic, *supra* note 11, at 567.

[33] *See* Christol, *supra* note 4, at 411.

[34] 1979 Principles, *supra* n. 20, Principle I.

[35] 1979 Principles, *supra* n. 20, Principle II.

[36] 1979 Principles, *supra* n. 20, Principle III.

[37] *See* 1979 Principles, *supra* n. 20, Principles IV, VI, VII.

[38] 1979 Principles, *supra* n. 20, Principle XIII.

[39] *Id.*

the earth by satellite and does not amount to an intolerable restriction on sensing States.[40] Furthermore, the need for such a provision is questionable in light of proposed Principle VII (2), Article XI of the Principles Treaty and Article IV of the Registration Convention, which require States to inform the Secretary-General of their space activities. As the Secretary-General makes this information public, the sensed State would have knowledge of the sensing in any event. It is reasonable to assume, however, that the "advance notification" of Principle XIII would be more particular and, more importantly, in "advance", something which is not assured under other provisions. The U.S. Working Paper of 1979 called for advance notification of remote sensing programs, to the extent feasible and practicable, to be given to the Secretary-General and the informing of other States as soon as practicable on their request.[41]

Principle XVI also touches on the right of a State to sense. It states that, "without prejudice to the principle of the freedom of exploration and use of outer space. . .remote sensing of the earth (should) (shall) be conducted with respect for the principle of full and permanent sovereignty of all States. . .over their natural resources."[42]

The right of a State to sense is also conditioned by proposed Principle XIV, which provides for consultations between the sensing and sensed State.

Thus, these appear to be few restrictions on the right of a State to conduct remote sensing of the earth and its environment under the 1979 Principles. It is to be hoped that no further limitations will be imposed.

C. The Right to Disseminate Primary Data and Analyzed Information

This issue is so explosive that the 1979 Legal Subcommittee did not even discuss it in order to avoid useless lengthy discussions. Divergent positions exist with respect to the right to disseminate primary data and analyzed information to third parties and the sensed State's right of access to the data and information. While some States favor universal dissemination, others maintain that the sensed States enjoy the exclusive right to receive the end product of the sensing process.[43] Complicating the matter is the fact that primary data and analyzed information are substantially different commodities.[44] While it may be argued that the raw data collection violates a State's sovereignty over its natural resources, it is necessary for the sensing State to process the data in order to convert it to useful information. The sensing State thus has substantial input in the analyzed information. Besides these distinctions, it is questionable whether sovereignty even extends to "information" concerning natural resources in the first place.[45] In any

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[40]Vlasic, *supra* note 11, at 570.

[41]U.N. Doc. A/AC.105/240, Annex I at 13, WG. III (1979) (W.P. 7).

[42]Note that many of the Principles are subject to a variety of different formulations.

[43]Christol, *supra* note 4, at 394.

[44]*Id.*

[45]*See* Cheng, *supra* note 2, at 605, para. 2.

event, the views of States on access to and dissemination of data and information closely parallel their views on the right to sense.

Again the better view is that of the United States, *i.e.* free dissemination of data and information. The free availability of facts advances world security and the better use of world resources. Global benefits are realized through the sharing of data and information. The aforementioned Presidential Directive of June 20, 1978 stated that "data and results from the civil space programs will be provided the widest practical dissemination to improve the condition of human beings on earth and to provide improved space services for the United States and other nations of the world". This policy is consistent with the United States Freedom of Information Act which requires that U.S. citizens have access to sensed data. The United States is opposed to the veto by individual States of the dissemination of data and information relating to their territories for a variety of reasons, including: (1) the aforementioned fact that satellites are not able to detect political boundaries; (2) the problems to which remote sensing satellites are addressed are of global, or at least regional, proportion; (3) it is unlikely that countries obtaining data could effectively operate ground stations under a restrictive dissemination system; and (4) a restrictive dissemination system would exacerbate the division between rich and poor countries and between technologically advanced and less advanced States.[46]

The 1979 Principles do provide an open regime for information relating to the preservation of the natural environment of the earth. Principle V calls for the protection of the environment and the necessity to make available information useful for the prevention of phenomena detrimental thereto. Principle VIII provides that data and/or information indicating an impending natural disaster shall be disseminated as promptly as possible to at least those States likely to be affected. There is general acceptance of both of these provisions.

Other Principles deal with data and information on a broad level. It is provided that the "data and information (shall) (should) be used by States in a manner compatible with the legitimate rights and interests of other States."[47] Also a consideration is the disputed provision calling for respect for the principle of sovereignty over wealth and natural resources, extending to the inalienable right to dispose of those resources and the information pertaining thereto.[48] Principle XIV calls for consultations between the sensing State and the sensed State in regard to the dissemination of data and information.

Besides the general principles, two conditional dissemination regimes are presented. The first accords to the sensed State timely and non-discriminatory access to data pertaining to its territory on reasonable terms. This access is to be on a continuous and priority basis and, in any case, no later than the dissemination of the data to any third State.[49] The second formulation is that States carrying out remote sensing of the

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[46]Leigh, *supra* note 30.

[47]1979 Principles, *supra* n. 20, Principle IX.

[48]1979 Principles, *supra* n. 20, Principle XVI.

[49]1979 Principles, *supra* n. 20, Principle XII.

earth shall not, without the approval of the States whose territories are affected by these activities, disseminate or dispose of any data or information on the natural resources of these States to third States, international organizations or public or private entities.[50] There is no general consensus on either of these versions.

One interesting proposal which did not find its way into the Principles was that of the U.S.S.R. which sought to classify the information acquired through remote sensing according to spatial resolutions. It suggested that there be "global," "regional" and "local" information. While global and regional information would be freely disseminated, local information, that with a resolution of 50 meters or finer, could not be distributed without the consent of the sensed State.[51] This classification, motivated by security considerations, was rejected.[52]

In summary, there is need for some legal regulation for the access of the sensed State to the data and information and of the right of the sensing State to distribute it to third parties. It can only be hoped that the limitations on an open, free system be kept to a minimum. As the social and economic benefits of open dissemination become known to those who now seek strict regulation, it is possible that many of the limitations now being suggested will have less appeal.[53] It is similar to legislating in advance for fear of some consequences that practice later shows to be unjustified, as, for example, the Concord case in New York.

III. International Cooperation

A recurring theme of the 1979 Principle is that of international cooperation, with special regard to be given to the developing countries. Both echoing and extending Article 1 of the 1967 Principles Treaty, the Principles speak of "international cooperation", the "benefit and. . .interests of all countries" and the "particular needs of the developing countries."[54] Principle IV calls on sensing States to promote international cooperation by making available to other States the opportunity to participate in their programs. States are also encouraged to consider agreements for the establishment of shared regional facilities. Principle VI advocates technical assistance. The United Nations and its agencies are also given a role to play. They should promote international cooperation, including technical assistance, and act as a coordinator of the remote sensing of the earth.[55] The United Nations is not only to be notified of remote sensing programs, but States are also encouraged to make available any relevant technical information involving possible operational systems which they are free to

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[50]1979 Principles, *supra* n. 20, Principle XV.

[51]See Christol, *supra* note 4, at 399 *et seq.*

[52]*Id.* at 417.

[53]*Id.* at 420.

[54]1979 Principles, *supra* n. 20, Principle II.

[55]1979 Principles, *supra* n. 20, Principle VII (1).

disclose.[56] Numerous other Principles call for agreement and consultation among States.[57]

Again it should be emphasized that many of these provisions are already found in other instruments of international law. It is also important to note that the transfer of technology problem and security considerations involved with the open dissemination of information will remain as obstacles to the acceptance by States of the Principles. Nevertheless, repeated and concentrated efforts to break down those barriers may prove successful in the long run.

Those States in favor of freedom of sensing and dissemination of data and information can point to the call for international cooperation as supportive of their position. Global and regional solutions to environmental and resource problems can only be achieved in a relatively open regime. Restrictions on sensing and dissemination are not easily reconciled with international cooperation. For there to be cooperation, there must be access to the information.

The U.S. LANDSAT program is an outstanding example of how a free system lends itself to international cooperation. The construction of ground stations around the world, joint participation in specific scientific programs and the wide dissemination of information are all evidence of the benefits which can be realized.[58]

In closing, it should be stressed that the emphasis on international cooperation does not preclude remote sensing by private concerns. While States are reluctant to assume international responsibility for the *ground* segment of private sensing programs (evidenced by the lack of consensus for Principle XI), such problems should be resolved in the future. The United States has already made extensive studies on the future role of government and private industry in remote sensing and how the transition to private concerns could be accomplished.[59]

In conclusion it may be stated that remote sensing is an essential space activity which should be used freely for the benefit of all. Any form of cooperation, collaboration, partnership or common exploitation as in INTELSAT should be envisaged, not limitations or restrictions. It is good behavior and friendly relations that should inspire the guidelines; political stands should not dictate the interdiction.

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[56] 1979 Principles, *supra* n. 20, Principle X.

[57] *See* 1979 Principles, *supra* n. 20, Principles XIV and XVII.

[58] *See* Christol, *supra* note 4, at 381-382.

[59] Private Sector Involvement in Civil Space Remote Sensing (PSIS), June 15, 1979. (Draft prepared by an Interagency Task Force).